

Exercice

Déterminer le polynôme $P \in \mathbb{R}[X]$ tel que $(X^2 + 1)P'' - 6P = 0$ et $P(1) = 2$.

si $P = 0$, $0(1) = 2$ impossible $\Rightarrow P \neq 0 \Rightarrow \deg(P) \in \mathbb{N}$. (*)

On pose $m = \deg(P)$

$$\Rightarrow P = \underbrace{a_m}_{\neq 0} X^m + a_{m-1} X^{m-1} + \dots + a_2 X^2 + a_1 X + a_0$$

$$\Rightarrow P' = m a_m X^{m-1} + (m-1) a_{m-1} X^{m-2} + \dots + 2 a_2 X + a_1$$

$$\Rightarrow P'' = m(m-1) a_m X^{m-2} + \dots + 2 a_2$$

$$(*) \text{ donne: } (X^2 + 1) [m(m-1) a_m X^{m-2} + \dots + 2 a_2] - 6 [a_m X^m + \dots + a_0] = 0$$

$$\Rightarrow m(m-1) a_m X^m - 6 a_m X^m + \dots + 2 a_2 - 6 a_0 = 0$$

$$\Rightarrow \boxed{(m(m-1) - 6) a_m} X^m + \dots + 2 a_2 - 6 a_0 = 0$$

$$\Rightarrow \underbrace{(m(m-1) - 6)}_{\neq 0} a_m = 0 \Rightarrow m(m-1) = 6 \Rightarrow \boxed{m = 3}$$

$$n=3 \Rightarrow P = aX^3 + bX^2 + cX + d$$

$$\Rightarrow P' = 3aX^2 + 2bX + c$$

$$\Rightarrow P'' = 6aX + 2b$$

$$(*) \text{ donne: } (X^2+1)[6aX+2b] - 6[aX^3+bX^2+cX+d] = 0$$

$$\Rightarrow \cancel{6aX^3} + 2bX^2 + 6aX + 2b - \cancel{6aX^3} - 6bX^2 - 6cX - 6d = 0$$

$$\Rightarrow -4bX^2 + 6(a-c)X + 2b - 6d = 0$$

$$\Rightarrow \begin{cases} -4b = 0 \Rightarrow b = 0 \\ 6(a-c) = 0 \Rightarrow a = c \\ 2b - 6d = 0 \Rightarrow d = 0 \end{cases} \Rightarrow P = aX^3 + \underbrace{b}_{=0}X^2 + \underbrace{c}_aX + \underbrace{d}_{=0}$$
$$= aX^3 + aX$$
$$= a(X^3 + X)$$

$$\text{Or, } P(1) = 2 \Rightarrow a(1^3 + 1) = 2$$

$$\Rightarrow 2a = 2$$

$$\Rightarrow a = 1$$

$$\text{Donc, } \boxed{P = X^3 + X.}$$